PNAS RESEARCH ARTICLE | BIOLOGICAL SCIENCES

Machine-learning dissection of endogenous A-to-I RNA editing reveals substrate- and target-level determinants of predictability

Author list pending[a,1]

This manuscript was compiled on August 25, 2026

Endogenous adenosine-to-inosine RNA editing reflects at least three scales: the intrinsic competence of an RNA duplex, its realization and encounter with ADAR in a cell, and selection of an individual adenosine. Which of these scales contains recoverable information beyond current sequence–structure models is unknown. We used machine learning as a controlled instrument for information dissection, evaluating biological annotations, supervision, representation, transfer, capacity, and integration on 884 human inverted-Alu substrates while quarantining the published pair-disjoint test. Substrate-level editing burden was predictable, and inter-arm geometry added reproducible information beyond graph-derived summaries, consistent with variation in realization or engagement. The tested RBP, transcript, repeat, and bulk chromatin measurements did not resolve that underlying cellular state sufficiently to improve site prediction. Two actionable principles emerged. Sampling substrates before sites prevented site-rich duplexes from dominating learning, separately from the benefit of recovering additional high- and low-editing observations. A target/helix-partner-aligned representation recovered local selectivity diluted by whole-duplex pooling; an identical encoder shifted 96 nucleotides away failed. A fixed equal-logit combination of independently trained whole-duplex-only and local-augmented graph predictors outperformed a preregistered unified network in development. Frozen before test access, ADARPro improved AUROC, AUPRC, and F1 in all five reported benchmark endpoints; mean improvements were 0.0277, 0.0261, and 0.0253, with positive pair-clustered intervals throughout. Endogenous editing predictability is hierarchical: substrate-level propensity is detectable but incompletely resolved by bulk measurements, whereas substrate-aware learning and target-centered representation provide actionable information.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Significance

Why one apparently suitable adenosine is edited while another is not remains unclear. Using a pair-disjoint machine-learning framework, we separated information carried by the RNA duplex, substrate-level editing propensity, and the local environment of the target nucleotide. Editing burden showed a reproducible component associated with pair geometry, yet current tissue-average annotations did not resolve the underlying cellular state. Learning equally from RNA substrates and explicitly aligning each target with its helix partner produced the strongest gains. The study identifies both what present data can predict and the matched cellular measurements needed next.

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA), diversifying transcripts while preventing endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Nevertheless, editing is selective: nearby adenosines within the same predicted structure can differ sharply, and the same locus can vary across tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context contribute to this selectivity (13–17), but they do not specify whether a potential substrate forms, persists, localizes appropriately, or encounters the relevant ADAR state.

Recent structure-aware predictors, exemplified by AdarEdit, model intrinsic editability over complete inverted-Alu substrates (18). Its compact graph attention network encodes nucleotide identity, local and global sequence, predicted secondary structure, pairing topology, geometry, thermodynamics, and motifs. A strict benchmark assigns entire Alu pairs globally to train, validation, or test, avoiding the severe leakage that would occur if sites from one duplex crossed partitions. Residual errors could therefore reflect missing

Author affiliations: [a]Affiliations pending author confirmation

Author contributions pending author confirmation.
Competing-interest declaration pending author confirmation.
Corresponding author and e-mail pending author confirmation.

cellular context, underused supervision, pooling that dilutes the focal nucleotide, or an inadequate separation of substrate-level propensity from local selectivity.

These explanations cannot be distinguished by adding one more annotation or a larger network. A biologically relevant variable can be redundant with sequence, measured at the wrong cellular scale, or too noisy to improve prediction. An architectural gain can likewise arise from overweighting substrates with many labeled sites rather than from a better account of editing. We therefore asked a broader question: within measurements available to the benchmark, where does incremental predictive information for endogenous A-to-I editing reside?

We preserved the published benchmark and examined biological context, supervision, substrate-wide and target-local representations, cross-tissue transfer, RNA foundation models, hierarchical formulations, and two forms of integration. Machine learning served both as predictor and controlled assay: changing supervision, representation, transfer, or integration tested which biological scales contain recoverable information beyond the sequence–structure graph. Every discovery comparison used pair-disjoint development folds and pair-clustered uncertainty; the canonical test remained sealed until one model was frozen. The evidence resolved four principles. Whole-duplex sequence and structure encode intrinsic competence. Substrate-level editing burden contains a reproducible component consistent with realization or engagement, although current bulk context proxies do not resolve the underlying cellular state. A target and its helix partner retain local information that whole-duplex pooling loses. Finally, the duplex is the effective statistical unit for learning from correlated sites. ADARPro follows this hierarchy rather than increasing capacity indiscriminately.

Results

A controlled framework separates biological scales. The benchmark comprises four individual GTEx tissue endpoints—artery tibial, brain/cerebellum, liver, and skeletal muscle—and Combined, a pooled endpoint rather than a fifth biological tissue. EditingIndex values at least 15% define positives and values below 1% define negatives; intermediate observations are excluded from the canonical binary task. The fixed global split contains 566 training, 139 validation, and 179 test pairs with zero overlap (Fig. 1).

Within the training and validation pairs, 5 outer folds generated out-of-fold (OOF) predictions for hypothesis testing and model selection. Feature fitting, checkpoint selection, and threshold choice were isolated from each outer fold. The internal reference was always trained and evaluated under the same folds as its candidate. Its lower absolute score reflects reduced training data and is not a replacement for published AdarEdit performance. The supplied-checkpoint canonical test remained the authoritative benchmark and was accessed only after the final model, thresholds, weights, code, and hashes had been frozen.

The campaign mapped hypotheses onto a biological hierarchy (Fig. 1A). Intrinsic duplex properties and cell- or transcript-specific state jointly govern whether a substrate is available to editing; local sequence and pairing then help distinguish adenosines within it. Outcomes were classified as selected, supported but not yet actionable, limited, not supported in the tested formulation, deferred for lack of matched measurements, or not pursued after stronger lower-complexity evidence emerged (Fig. 2A). These categories separate biological relevance from incremental utility at the resolution tested here.

The substrate is a biological and statistical unit. The canonical row set samples a subset of the available observations, whereas the underlying training tables contain additional sites with valid high or low EditingIndex. Recovering these labels improves supervision, but uniform site sampling gives a substrate weight proportional to its number of observations. Sites within one duplex share sequence, structure, genomic context, and often editing behavior; they are not independent biological units.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 3A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair a further +2.27 AUROC points (+1.25 to +3 to +4.01). The two corrections therefore ma

The combined pair-first formulation impro five endpoints (Fig. 3B). All 15 AUROC, AU zero, with F1 gains from +0.69 to +4.23 per result rather than a claim that the pair is a sin pair-first sampling corrects the effective cont

The substrate also carries biological varia separation between Alu arms—information c them with a fixed separator—correlated neg in every individual tissue (Spearman ρ = − capacity models combining geometry with gra edited fraction with R^2 = 0.050–0.277 and alone in every tissue (Fig. 2C; SI Appendix a reproducible substrate-level component of variation in realization or engagement but no process.

Bulk context does not resolve the latent cellular state. actionable for site prediction proved hard fitted engagement gate added 0.0084 AUP interval included zero, and increased Brie engagement auxiliary likewise failed its prespe geometry marks a stable propensity; it can transcript formed the expected duplex, remai compartment, or encountered active ADAR.

The initial cellular-context hypothesis fo combined replicated ENCODE eCLIP occu predictions, literature-prioritized regulators, by-abundance interactions, and randomize Alu multimapping makes a missing eCLIP p predictors supplied a complementary test inde cell-line occupancy nor tissue-average abun residual information. Cross-tissue label switc tissue identity alone, preventing four tissue specific transferable mechanism. These mea to resolve transcript-specific regulator expos assays are the more informative next test (2

Broad transcript, repeat-subfamily, diverg tions likewise failed to improve site-level pre tissue bulk H3K36me3 overlap produced a without stable AUROC, F1, or pair-level bene resource supported a powered test. The we toward quantitative, sample-matched assays than binary tissue-level overlap (27, 28). To measurement problem: substrate-level prope static and tissue-average proxies do not res improve the strongest site classifier.

Target alignment resolves site-level selectivity. The a strong account of whole-duplex competi embeddings can dilute the state of one targe localized the missing information. Simple ta ranking and calibration but not F1, and exp node improved AUROC and AUPRC more matched duplicate-global control. Both results a specialized local encoder was built.

The selected local-augmented graph pre sentation centered jointly on the focal adeno helix partner. A low-capacity dilated convolu pairing state, focal and partner indicators, output is fused with the whole-duplex graph In muscle OOF data it reached AUROC AU

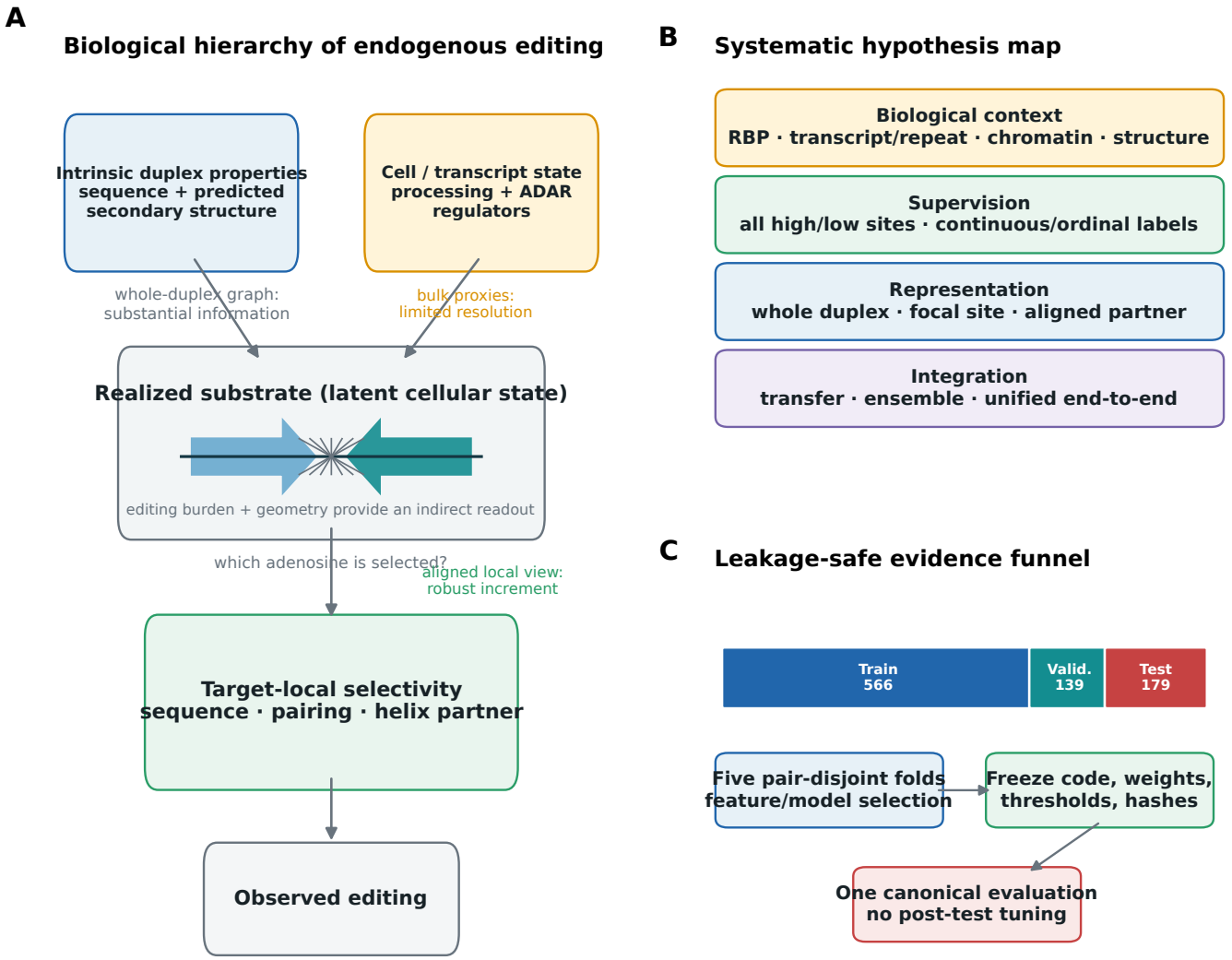


Fig. 1. A biological hierarchy and leakage-safe dissection of editing predictability. (A) Intrinsic duplex properties and cell/transcript state converge on whether an inverted-Alu substrate is available to editing; target/helix-partner context then contributes to selection of a particular adenosine. Pair-level editing burden provides an indirect readout of substrate propensity, whereas the underlying cellular realization state was not directly measured. (B) Controlled AI interventions tested biological context, supervision, representation, and integration. (C) Whole pairs were partitioned globally. All discovery used pair-disjoint development folds; the final specification was frozen before one canonical held-out evaluation. Combined was never assigned a fabricated tissue state.

compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 3C).

The decisive control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. The matched-capacity shift collapses performance, localizing the incremental information to the target-centered representation rather than to added convolutional capacity. Replication in liver and secondary natural-prevalence analysis supported the same direction.

Whole-duplex and target-local information are complementary. The local-augmented predictor did not make an independently optimized whole-duplex view redundant. Their errors remained sufficiently distinct that a fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver; all six pair-clustered intervals excluded zero (Fig. 3D).

ADARPro was consequently specified as two independently trained pair-first components: a whole-duplex-only graph predictor and a second graph predictor internally augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed, unsearched 0.5 tests whether independently optimized glob remain complementary without fitting a flex independent substrates.

Additional complexity provides limited returns. In contain graded information, but most useful recovered by pair-first binary training. A bin head increased AUROC by 0.0081 (95% pair while changing F1 by only 0.0008; its integr prevalence. Shared four-tissue training likewis mean changes across the four individual tiss 0.0114 AUPRC, and 0.0075 F1, but missed i and worsened mean Brier score by 0.0043. sharing therefore refine ranking, but less than representation.

Other increases in capacity were unprodu trained all candidate adenosines jointly chang -0.0599 relative to the pair-first model and me boundary. Frozen RiNALMo target and s

A

Evidence map: what the benchmark can and cannot currently resolve

Scale	Tested formulation	Status	Interpretation
Substrate-aware learning	Pair-first expanded supervision	Selected	Corrects label use and pair weighting
Target selectivity	Aligned target + helix-partner view	Selected	Survives shifted-window control
Integration	Fixed whole-duplex + local-augmented logits	Selected	Complementary same-fold predictions
Substrate propensity	Pair geometry / editing burden	Supported*	Editing burden is learnable; latent state unresolved
Supervision	Continuous + ordinal auxiliary labels	Limited	Ranking benefit; no useful F1 gain
Target selectivity	Focal node / contact geometry	Limited	Points toward aligned local view
Cross-tissue	Shared four-tissue training	Limited	Transfer below promotion criterion
Substrate propensity	Hierarchical engagement gate	Limited	Small ranking gain; unstable classification
Biological context	Bulk H3K36me3 overlap	Limited	Context-specific AUPRC only
Biological context	RBP binding × bulk expression	Not supported	No reproducible residual gain
Biological context	Transcript / repeat annotation panel	Not supported	No robust site-level increment
Representation	Frozen RNA-language embeddings	Not supported	No consistent incremental utility
Representation	Substrate-wide all-A decoder	Not supported	Stopped at prespecified futility gate
Capacity	Larger / deeper graph networks	Not supported	Capacity alone did not generalize
Integration	Unified end-to-end architecture	Not supported	Inferior to frozen ensemble
Cellular state	Matched donor/cell multimodality	Deferred	Required measurements unavailable
RNA state	Matched in vivo structure / isoforms	Deferred	Required measurements unavailable
Capacity	Graph Transformer / GATv2 expansion	Not pursued	Superseded by controlled lower-capacity evidence

Status reflects prespecified same-fold comparisons.
*Reproducible observation not yet actionable for site prediction; deferred hypotheses were not tested.

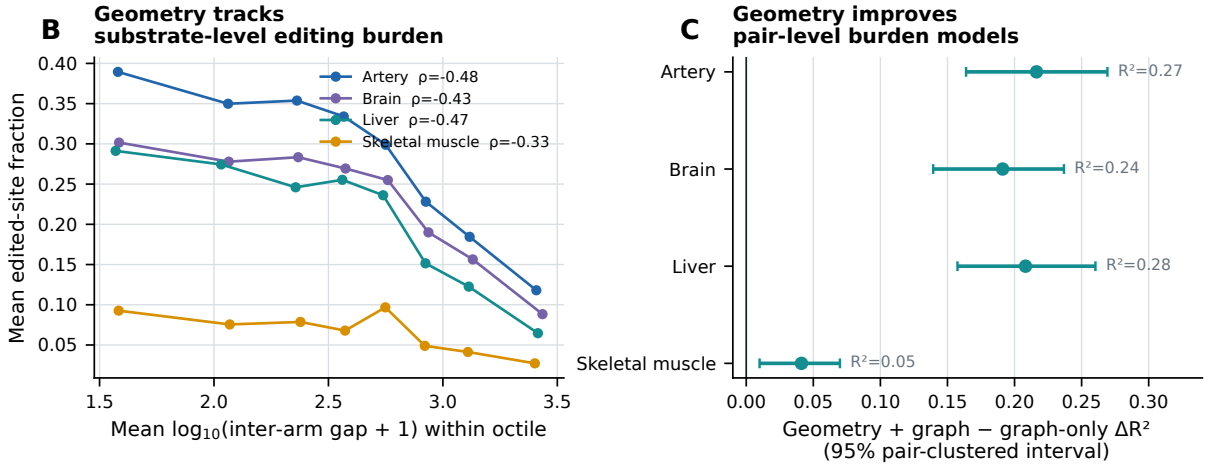


Fig. 2. A structured evidence map and positive evidence for substrate-level propensity. (A) Hypotheses are grouped by biological or learning scale and epistemic outcome. Status reflects prespecified same-fold comparisons and controls; unsupported refers only to the tested formulation, deferred measurements were unavailable, and superseded branches were not interpreted as negative. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 from adding genomic geometry to graph-derived summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

a larger model, added little beyond the task-specific structure graph (29). Larger or deeper graph networks likewise failed stronger-model controls. More parameters were not a substitute for the correct information structure. The more pertinent pretraining question—benchmark-safe, inverted-Alu-specific representation learning—remained deferred because a leakage-safe corpus could not be guaranteed.

Before test access, we preregistered one connected architecture to ask whether the surviving principles interact more effectively during joint optimization. It combined a three-layer graph encoder, focal-node state, aligned local encoder, pairwise representation interactions, and a training-only substrate head. Its 65,698 trainable parameters were fewer than the 80,138 parameters across the two ensemble components. On the 5,967-row, 585-pair muscle development set, the unified model reached 0.9072/0.8922/0.8317 AUROC/AUPRC/F1 versus 0.9142/0.9040/0.8347 for the frozen ensemble. AUROC and AUPRC were worse in every fold; their pair-clustered int
Brier increased by 0.0205 (0.0135–0.0276). P
branches remained closed. The empirical c
joint architecture did not outperform fixed p
effective sample size.

A predictor built from the hierarchy improves every can
selection, 10 component fits were trained on p
validation data alone selected checkpoints, epo
per endpoint. The exact code revision, con
were bound into a final manifest. A single
canonical test for one evaluation; no mode
changed afterward.

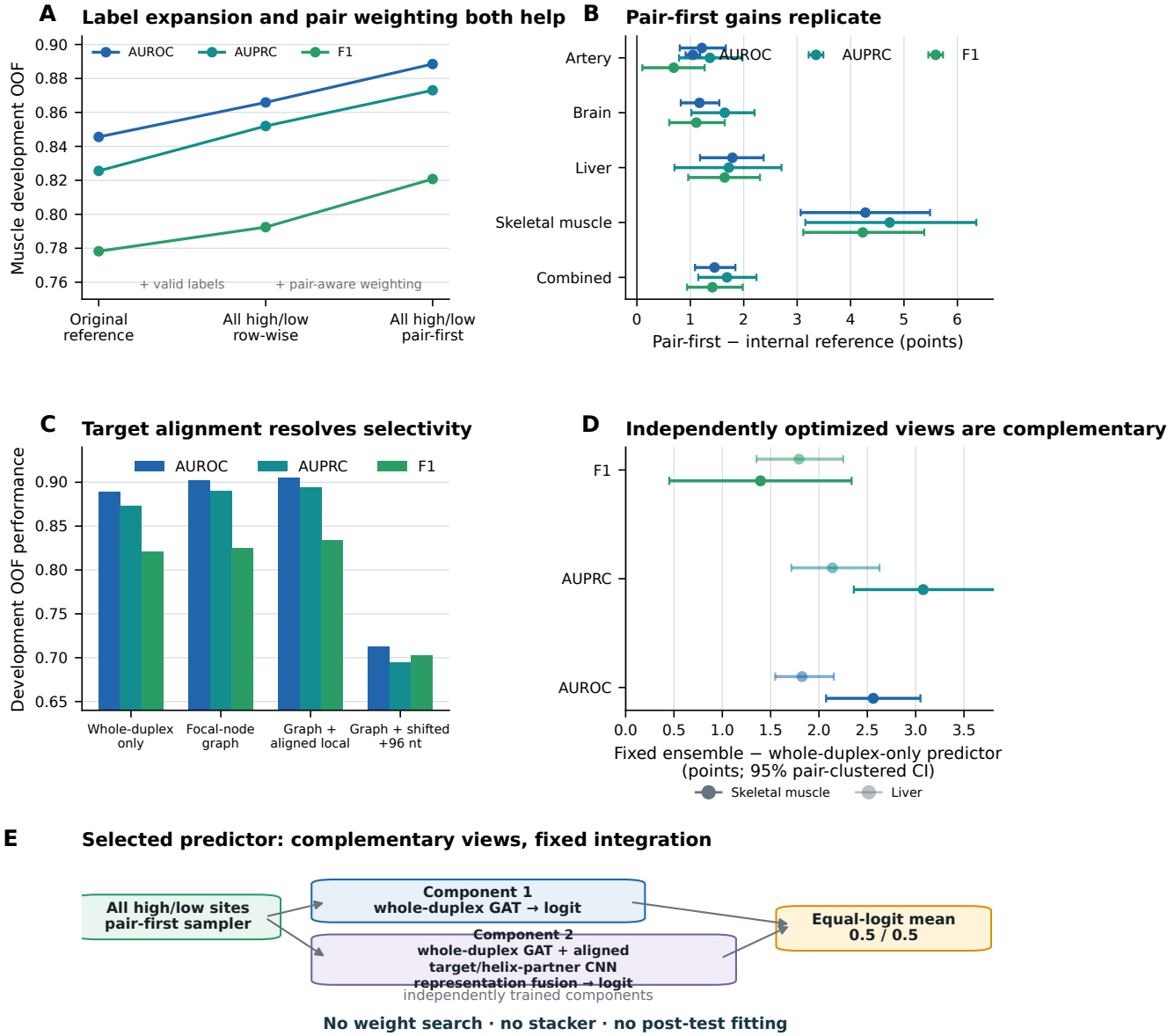


Fig. 3. The two actionable AI principles and their integration. (A) Same-fold ablations separate the benefit of recovering additional valid high/low labels under row-wise sampling from the further benefit of pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, explicit focal-state retention, and graph-plus-aligned-local fusion form an evidence progression; an otherwise identical local-augmented model shifted by 96 nt fails. (D) The fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) Exact selected predictor: an independently trained whole-duplex-only graph model and graph-plus-local fusion model, followed by a fixed 0.5/0.5 logit mean. No ensemble weight, stacker, or post-test parameter is fitted. All panels show controlled development evidence.

ADARPro improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 4A). Mean gains across those endpoints over reproduced supplied-checkpoint AdarEdit were 0.0277, 0.0261, and 0.0253. Because Combined overlaps the individual tissue endpoints, this mean is descriptive rather than a five-tissue meta-analysis. F1 changed from 0.8671 to 0.8869 in artery, 0.8689 to 0.8902 in brain, 0.8541 to 0.8812 in liver, 0.8139 to 0.8541 in skeletal muscle, and 0.8674 to 0.8852 in Combined. Every endpoint had a positive 95% pair-clustered bootstrap interval for all three metrics, and Brier score decreased throughout (Fig. 4C).

The gains were not purchased by indiscriminate suppression of positive calls. At frozen validation-selected thresholds, ADARPro corrected 32.0–50.6% of baseline false positives while damaging 1.7–5.9% of baseline true positives (Fig. 4D). In a post-test descriptive analysis that rethresholded ADARPro to match AdarEdit recall exactly, the point estimates were 126, 119, 158, 68, and 148 fewer false positives across arte and Combined; every pair-clustered interva canonical gain provides an out-of-sample test the hierarchy inferred entirely in developmen

Discussion

This hierarchy describes recoverable predic complete causal decomposition of RNA-edi captures substantial intrinsic duplex comp nevertheless contains an additional substr geometry covaries with editing burden, and that burden beyond graph-only summaries substrate, the fate of an adenosine retains targ global pooling dilutes. Sites sharing a duplex

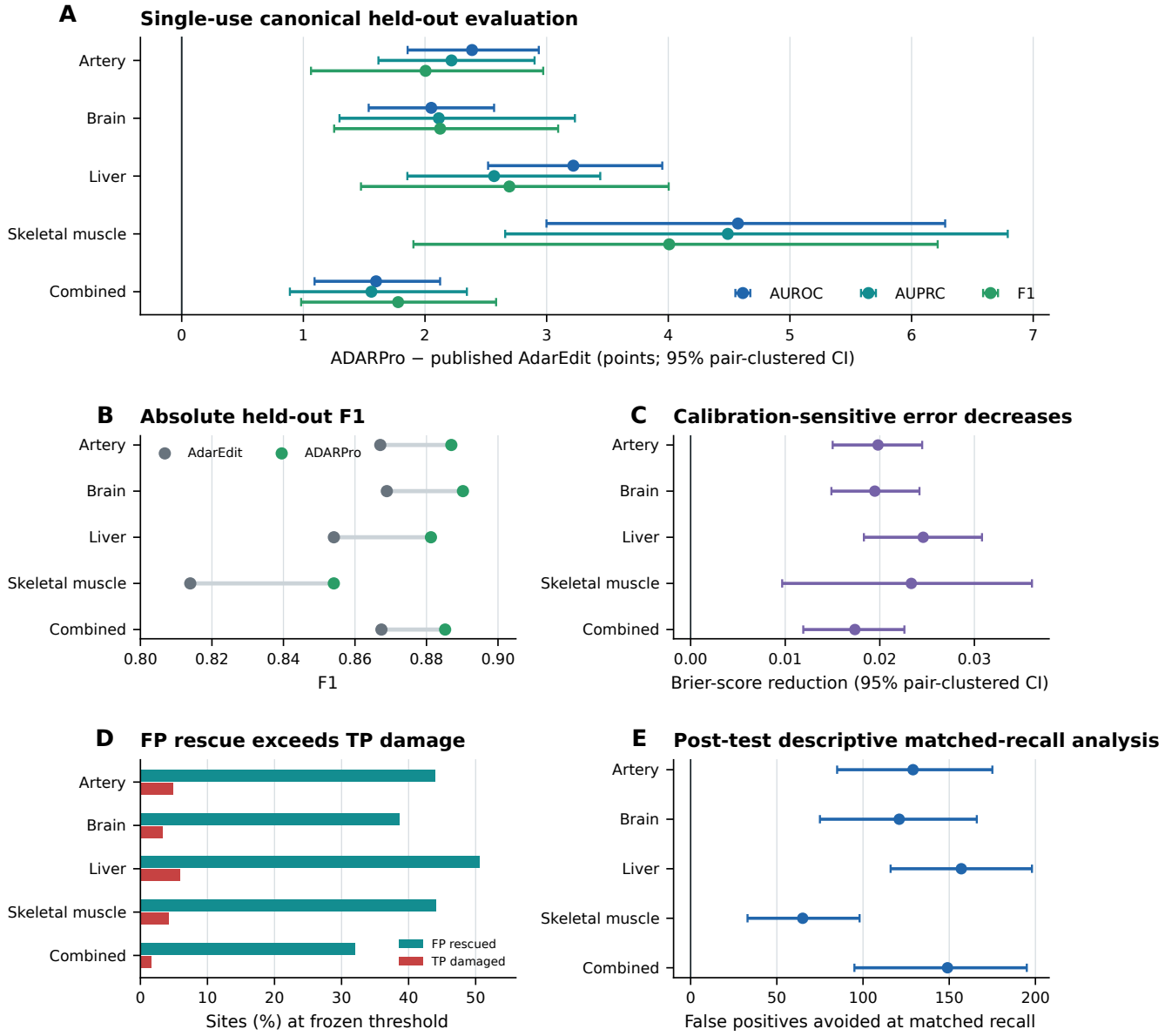


Fig. 4. Single-use canonical held-out evaluation. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score, a calibration-sensitive error measure. (D) Baseline false positives corrected and baseline true positives damaged at frozen validation-selected thresholds. (E) Descriptive post-test false-positive reduction after rethresholding ADARPro to match AdarEdit recall. Combined is a pooled endpoint. All quantities were generated after the final manifest was frozen and the one-use authorization was consumed.

unit; learning improves when substrates, rather than abundant rows, control effective weight.

These levels should not be conflated. Pair-first sampling is a statistical correction motivated by biological dependence, not evidence that all sites in a substrate share one causal fate. Conversely, the pair-level findings are positive biological observations even though their integration did not improve the leading site classifier. Geometry estimates a stable propensity. Actual realization and engagement also depend on transcription, splicing, residence time, localization, in vivo conformation, ADAR isoform and activity, and regulator occupancy—states averaged away by tissue medians and binary peaks.

The limited and unsupported branches locate the remaining measurement gap. Cell-line RBP occupancy combined with tissue-median abundance supplied no reproducible residual information about transcript-specific regulator exposure. The weak bulk H3K36me3 effect makes quantitative, sample-matched cotranscriptional measurements the sharper next test. Cross-tissue sharing and continuous labels yielded small ranking
and graded structure exists but contributes le
target alignment. Generic frozen RNA embed
likewise did not substitute for information lo

The preregistered unified experiment quali
representation-level integration is not automat
level integration. The tested connected archit
well than the frozen ensemble, despite incorp
using fewer total parameters. The comparison
mechanism. It does establish why the final p
whole-duplex-only graph and an independent
combined without a learned weight.

Several limitations define the scope of t
contains graph-buildable human inverted-Alu
not donor-resolved measurements or all editin
structure is a surrogate for the conformations

canonical task excludes intermediate observations and is balanced; natural-prevalence analyses are secondary. eCLIP coverage is cell-line-specific and compromised by repeat mapping. Finally, the canonical partition is stringent but derives from the same benchmark construction; validation in an independent cohort remains necessary.

The canonical gain provides an out-of-sample test of the predictive consequences of this hierarchy. Better prediction followed from respecting substrate dependence and preserving target-centered context, not from indiscriminate feature or capacity growth. The major unresolved layer is the cellular state underlying substrate-level propensity. The next decisive study should measure editing, transcript abundance and processing, splicing and intron retention, ADAR1 isoform and interferon state, ADAR activity and localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation in the same donors or cells and on the same substrates. Such data could directly test a two-stage model in which cellular state governs whether an intrinsically competent substrate is realized or engaged, and local RNA context governs which adenosine is selected.

Materials and Methods

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human labels were edited for EditingIndex $\geq 15\%$, nonedited for EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. Pair identities and the fixed 566/139/179 train/validation/test assignment were verified before analysis. Combined was treated only as its published pooled endpoint.

Development protocol. All discovery used the original train plus validation pairs divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice were fitted without the outer fold. Predictions were concatenated only after every row was scored by a model that had not trained on its pair. The test paths were protected by a stateful quarantine. Development OOF metrics estimate same-fold incrementa remain the authoritative reference.

Models. The first component retained the compact Ada from the expanded high/low site pool using uniform p second, independently trained component retained the position target/helix-partner-aligned representation enco local and graph embeddings were fused before its predi $\sigma[(\operatorname{logit} p_{\text{whole}} + \operatorname{logit} p_{\text{augmented}})/2]$; weights were fi shifted-window control preserved the graph-plus-convolution the local center 96 nt. Full feature definitions, architectures are in SI Appendix.

Statistics and final evaluation. Metrics were AUROC, balanced accuracy, MCC, and Brier score. Thresholds m training run. Uncertainty used a pair-clustered bootstrap wi and retained all their sites. For the post-test descriptive mato was selected on final predictions solely to equal supplied-c resampled pairs. Before final evaluation, code, inputs, we SHA-256 values were frozen. Test authorization was consu or tuning occurred. Natural-prevalence results use all eligib pairs under unchanged pair assignments and are seconda

Data, Materials, and Software Availability. Code, experime figure scripts, retained derived artifacts, frozen checkpoin //github.com/Scientific-Computing-Lab/ADARPro. Third-p omitted from GitHub are listed with source URLs, roles, che data manifest. The AdarEdit benchmark is available from h AdarEdit. No new human participants were recruited ar generated.

ACKNOWLEDGMENTS. Acknowledgments and funding info

1. B. L. Bass, Rna editing by adenosine deaminases that act on rna. *Annu. Rev. Biochem.* **71**, 817–846 (2002).
2. K. Nishikura, A-to-I editing of coding and non-coding RNAs by ADARs. *Nat. Rev. Mol. Cell Biol.* **17**, 83–96 (2016).
3. E. Eisenberg, E. Y. Levanon, A-to-I RNA editing—immune protector and transcriptome diversifier. *Nat. Rev. Genet.* **19**, 473–490 (2018).
4. B. J. Liddicoat, et al., RNA editing by ADAR1 prevents MDA5 sensing of endogenous dsRNA as nonself. *Science* **349**, 1115–1120 (2015).
5. N. M. Mannion, et al., The RNA-editing enzyme ADAR1 controls innate immune responses to RNA. *Cell Reports* **9**, 1482–1494 (2014).
6. E. Y. Levanon, et al., Systematic identification of abundant A-to-I editing sites in the human transcriptome. *Nat. Biotechnol.* **22**, 1001–1005 (2004).
7. A. Athanasiadis, A. Rich, S. Maas, Widespread A-to-I RNA editing of Alu-containing mRNAs in the human transcriptome. *PLoS Biol.* **2**, e391 (2004).
8. D. D. Y. Kim, et al., Widespread RNA editing of embedded Alu elements in the human transcriptome. *Genome Res.* **14**, 1719–1725 (2004).
9. L. Bazak, et al., A-to-I RNA editing occurs at over a hundred million genomic sites, located in a majority of human genes. *Genome Res.* **24**, 365–376 (2014).
10. H. T. Porath, S. Carmi, E. Y. Levanon, A genome-wide map of hyper-edited RNA reveals numerous new sites. *Nat. Commun.* **5**, 4726 (2014).
11. M. H. Tan, et al., Dynamic landscape and regulation of RNA editing in mammals. *Nature* **550**, 249–254 (2017).
12. E. Picardi, et al., Profiling RNA editing in human tissues: towards the inosinome atlas. *Sci. Reports* **5**, 14941 (2015).
13. J. M. Eggington, T. Greene, B. L. Bass, Predicting sites of ADAR editing in double-stranded RNA. *Nat. Commun.* **2**, 319 (2011).
14. S. K. Wong, S. Sato, D. W. Lazinski, Substrate recognition by ADAR1 and ADAR2. *RNA* **7**, 846–858 (2001).
15. K. A. Lehmann, B. L. Bass, Double-stranded RNA adenosine deaminases ADAR1 and ADAR2 have overlapping specificities. *Biochemistry* **39**, 12875–12884 (2000).
16. L. Bazak, E. Y. Levanon, E. Eisenberg, Genome-wide analysis of Alu ed
17. M. S. Zambrano-Mila, et al., Dissecting the basis for differential substrate 8212 (2023).
18. Z. Rosenwasser, M. Levitt, E. Y. Levanon, G. Oren, Structure-aware grap species (bioRxiv preprint) (2026).
19. E. L. Van Nostrand, et al., A large-scale binding and functional map of hu
20. L. Sun, et al., Predicting dynamic cellular protein–RNA interactions by de 495–516 (2021).
21. M. Horlacher, et al., Towards in silico CLIP-seq: predicting protein–RNA **24**, 180 (2023).
22. GTEx Consortium, The GTEx Consortium atlas of genetic regulatory eff (2020).
23. E. C. Freund, et al., Unbiased identification of trans regulators of ADAR
24. G. Quinones-Valdez, et al., Regulation of RNA editing by RNA-binding p
25. E. J. Park, et al., Primate lineage specification requires suppression of A
26. Y. Yi, et al., A dual role of EZH2 in regulating A-to-I RNA editing and mRN
27. P. Cai, et al., Chromatin state shapes site-specific A-to-I RNA editing (bi
28. ENCODE Project Consortium, Expanded encyclopaedias of DNA eleme 699–710 (2020).
29. R. J. Penic, T. Vlasic, R. G. Huber, Y. Wan, M. Sikic, RiNALMo: general- structure prediction tasks. *Nat. Commun.* **16**, 5671 (2025).